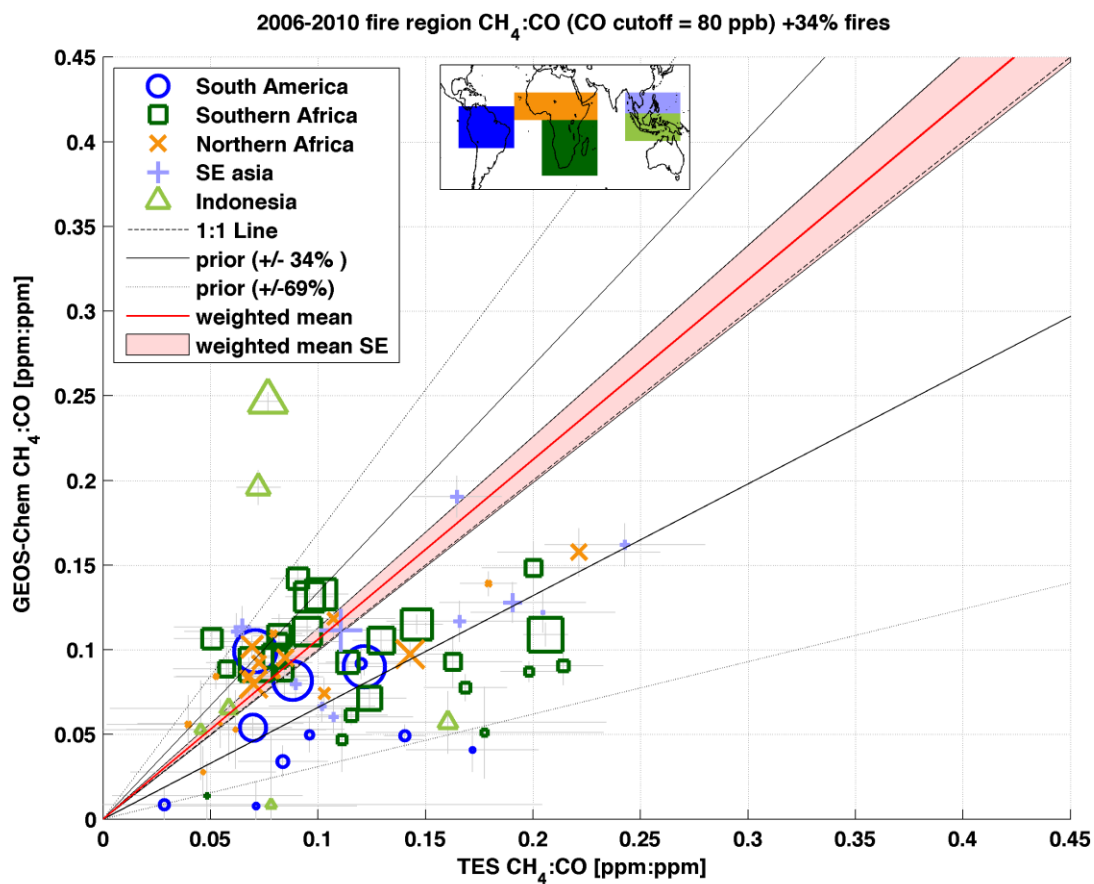
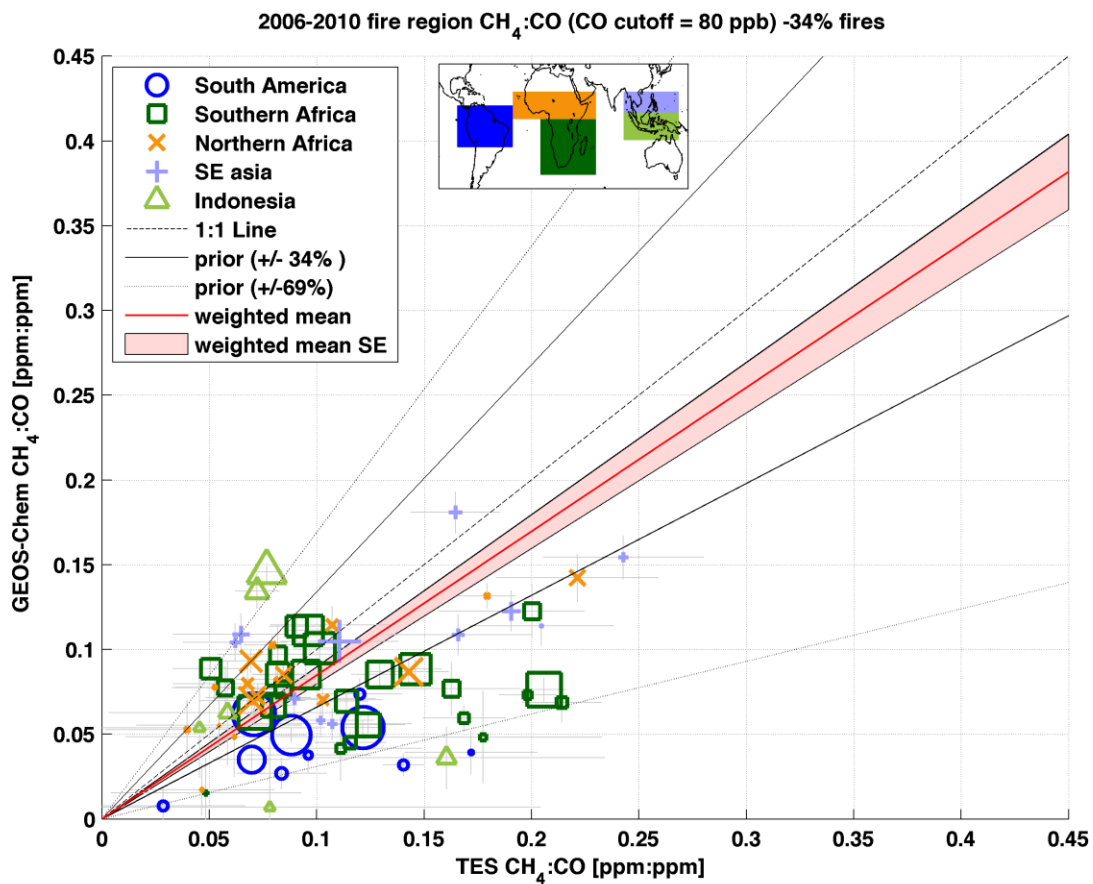
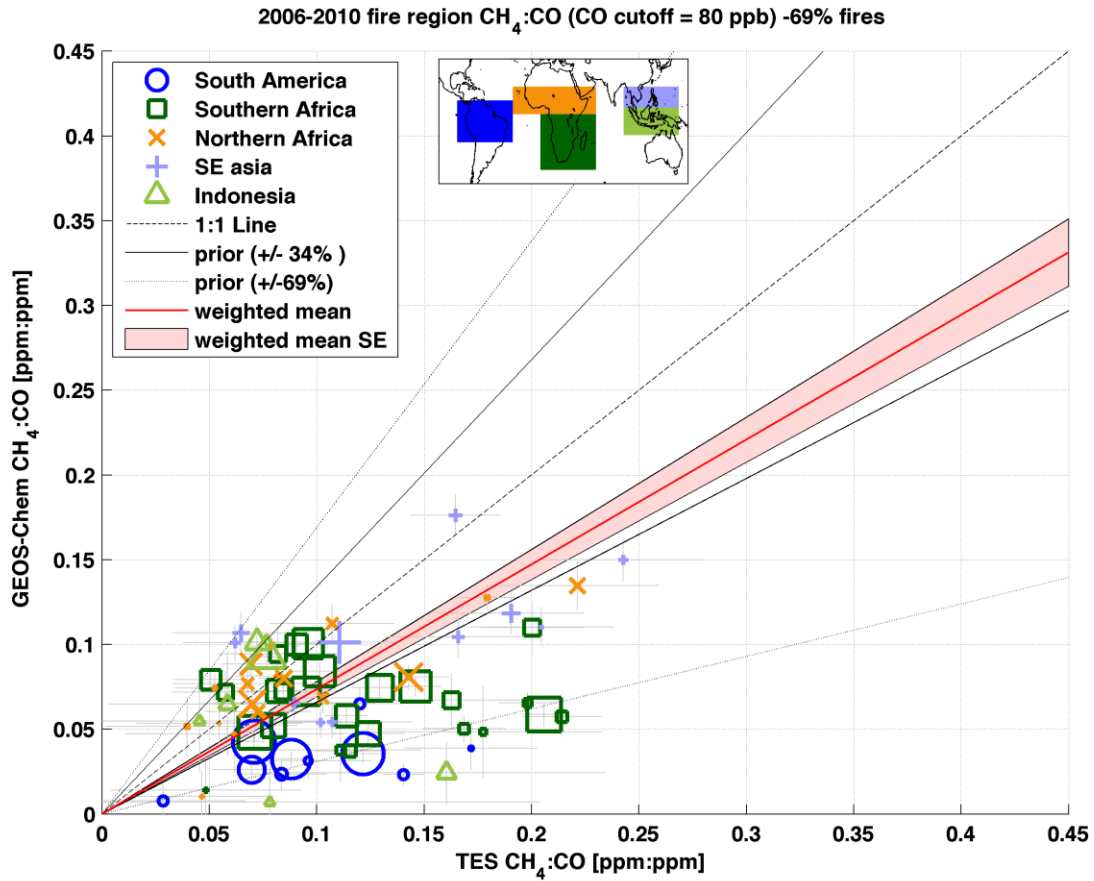




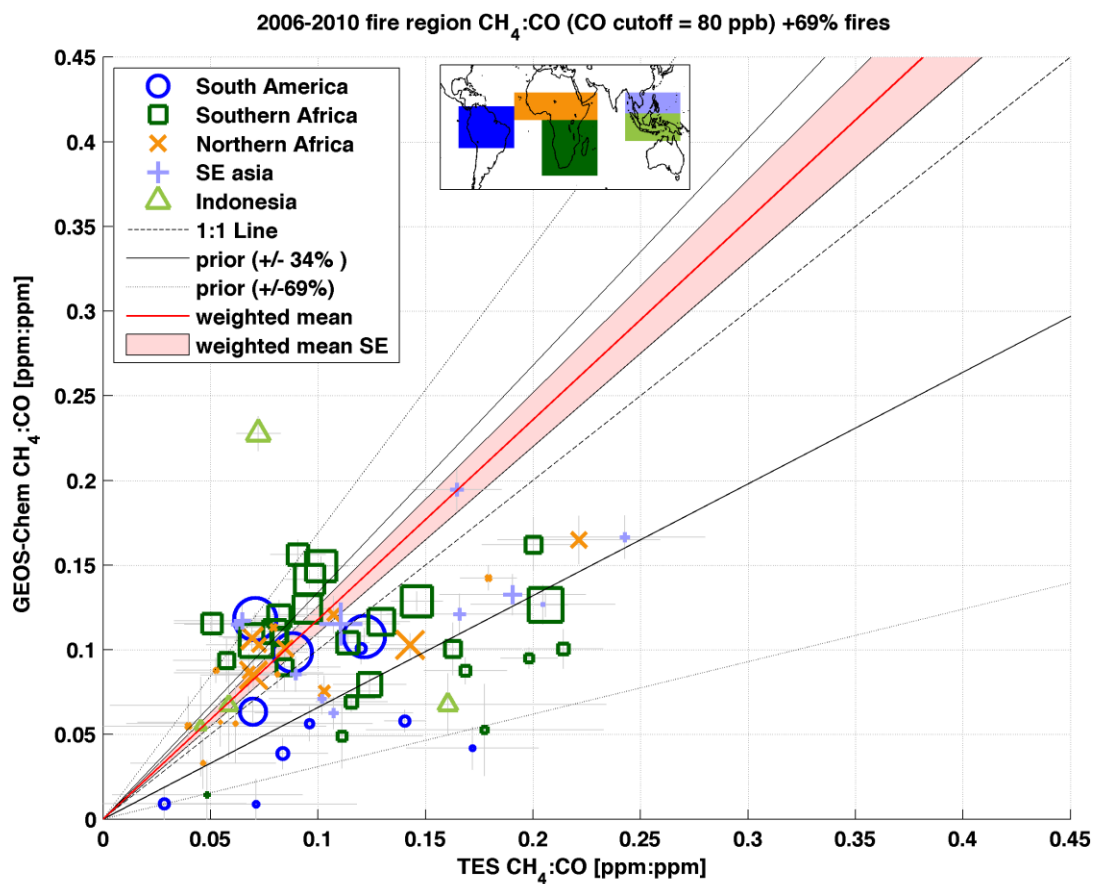
Supplementary Figure 1: Comparison Of CH_4 /CO ratios from the GEOS-Chem model and Aura TES data. This figure is the same format as in Figure 6 but the CH_4 /CO ratios in the model have been changed by 34%



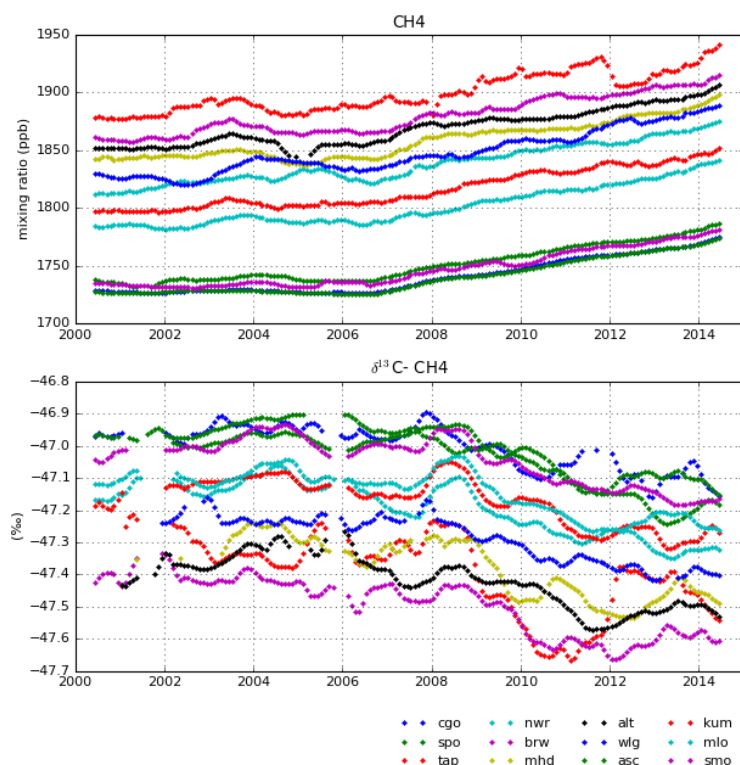
Supplementary Figure 2: Comparison Of CH_4 /CO ratios from the GEOS-Chem model and Aura TES data. This figure is the same format as in Figure 6 but the CH_4 /CO ratios in the model have been changed by -34%.



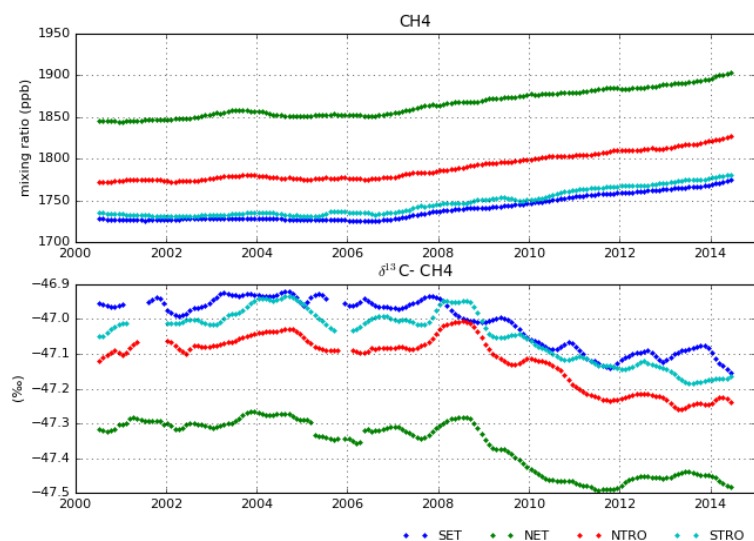
Supplementary Figure 3: Comparison Of CH_4/CO ratios from the GEOS-Chem model and Aura TES data. This figure is the same format as in Figure 6 but the CH_4/CO ratios in the model have been changed by -69%.



Supplementary Figure 4: Comparison Of CH_4 /CO ratios from the GEOS-Chem model and Aura TES data. This figure is the same format as in Figure 6 but the CH_4 /CO ratios in the model have been changed by +69%.



Supplementary Figure 5: CH₄ and δ¹³C-CH₄ measurements smoothed using a 12-month running mean. (Panel a) CH₄ measurements for all stations used in this analysis. The different colors correspond to the data from the different stations used in this analysis. (Panel b) Similar to panel a but for the corresponding δ¹³C-CH₄ measurements. The different stations are Ascension island (asc), Tae-ahn (tap), Cape Grim (cgo), Niwot Ridge (nwr), Cape Kumukahi (kum), Barrow (brw), Mauna Loa (mlo), Mace Head (mhd), Tutuila (smo), Alert (alt), the South Pole (spo), and Mt. Waliguan (wlg).



Supplementary Figure 6: Zonal means of the measurements shown in Figure 11.
 (Panel a) Zonal means of CH₄. (Panel b) Zonal means of δ¹³C-CH₄. NET (Northern Extra Tropics: 30 N to 90 N), NTRO (Northern Tropics: 0 to 30 N), STRO (Southern Tropics: 30 S to 0), SET (Southern Extra Tropics: 90 S to 30 S)

Supplementary Table 1: Assessment of the optimal starting year for the flux inversion time window. The goodness of fit is measured by the RMSD between CH₄ mole fractions measurements and optimized box model simulations for starting years varying between 2005 and 2008. The corresponding strength of the optimized methane emission perturbation is also given.

Years of perturbation	RMSD (ppb)	Strength of emission perturbation (TgCH ₄ /yr)
2005	5.0	18.32
2007	1.7	24.7
2006	2.8	21.2
2008	4.0	28.4